POISONING

Forensic Medicine | 3rd Year MBBS

Hi Sammy!
Poisoning = MOST asked topic in Toxicology!
Learn antidotes cold!

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Poison = any substance that causes harm when administered. IPC is key!

DEFINITION

"A substance which when introduced into or applied to the body can injure health or destroy life." - Reddy

Classification of Poisons

Type	Examples
Corrosive	Strong acids (H_2SO_4, HNO_3, HCl), Strong alkalis (NaOH, KOH)
Irritant — Organic	Castor, croton, abrus, calotropis, oleander
Irritant — Inorganic	Arsenic, mercury, lead, copper, phosphorus
Systemic — CNS	Opium, barbiturates, alcohol, dhatura, cannabis
Systemic — CVS	Aconite, digitalis, oleander
Systemic — Spinal	Strychnos nux vomica (strychnine)
Asphyxiants	CO, CO_2, H_2S, cyanide (HCN)
Miscellaneous	Snake venom, insecticides (OP, carbamates)

IPC Sections Related to Poisoning

Section	Provision
Sec 284	Negligent conduct with poison — endangering life
Sec 299/300	Culpable homicide / Murder by poisoning
Sec 304A	Death by negligence (accidental poisoning)
Sec 326	Grievous hurt by poison (voluntary)
Sec 328	Administering poison to commit offence

Factors Modifying Action of Poison

- Dose & concentration | Route of administration (IV > inhalation > oral > topical)
- Age (children more susceptible) | Habit & tolerance (chronic alcoholics)
- State of health (liver/kidney disease) | Idiosyncrasy & hypersensitivity
- Form of poison (liquid > solid) | Full vs empty stomach (absorption speed)

OP poisoning = THE
most asked topic!
Learn DUMBELS mnemonic!

Organophosphorus (OP) Poisoning

Mechanism: OP inhibits Acetylcholinesterase (AChE) irreversibly
-> Accumulation of Acetylcholine -> Continuous stimulation of receptors
Examples: Malathion, Parathion, Diazinon, Chlorpyrifos (Tik-20)

★ **DUMBELS — Muscarinic Effects**
D = Diarrhea | U = Urination | M = Miosis | B = Bradycardia/Bronchospasm
E = Emesis | L = Lacrimation | S = Salivation/Sweating

Nicotinic Effects: Muscle fasciculations, twitching, weakness, paralysis, tachycardia
CNS Effects: Anxiety, restlessness, seizures, coma, respiratory failure (cause of death)

Treatment of OP Poisoning

Step	Details
Decontamination	Remove clothes, wash skin with soap & water, gastric lavage
Atropine (main Rx)	Blocks muscarinic effects. Give till full atropinization
PAM / Pralidoxime	Reactivates AChE. Give within 24h BEFORE aging occurs
Supportive	O2, ventilator support, diazepam for seizures
Atropinization signs	Dry mouth, dilated pupils, tachycardia, flushed skin

Corrosive Acid Poisoning (Vitriolage)

Acid	Key Feature	PM Finding
Sulphuric (H_2SO_4)	King of poisons	Black vomit, charred mucosa
Nitric (HNO_3)	Yellow staining	Yellow mucosa (xanthoproteic)
Hydrochloric (HCl)	White precipitate	Grayish-white mucosa
Carbolic (Phenol)	Whitening of mucosa	Carboluria (smoky green urine)

★ **Alkali (NaOH/KOH): Soapy feel, transparent mucosa (saponification of fats),**
stricture of esophagus (late complication). Acids affect stomach; Alkalis affect esophagus!

Cherry red lividity = CO poisoning. Methanol vs Ethanol is classic Q!

Carbon Monoxide (CO) Poisoning

Feature	Details
Mechanism	CO binds Hb with 210x affinity -> Carboxyhaemoglobin (COHb)
Cherry-red color	COHb gives cherry-red lividity & blood color (PM sign)
Symptoms	Headache -> confusion -> coma -> death (at >60% COHb)
Treatment	100% O2 therapy, hyperbaric O2 in severe cases
PM finding	Cherry-red lividity, bright pink blood, pink organs

Methanol vs Ethanol Poisoning

Feature	Methanol (Wood Alcohol)	Ethanol (Grain Alcohol)
Toxic metabolite	Formaldehyde -> Formic acid	Acetaldehyde -> Acetic acid
Target organ	EYES (retina, optic nerve)	CNS (brain)
Key symptom	Blindness (snowstorm vision)	Inebriation, coma
Acidosis	Severe metabolic acidosis	Mild ketoacidosis
Treatment	Ethanol / Fomepizole + dialysis	Supportive, thiamine
Fatal dose	60-240 mL	150-250 mL (absolute)

Heavy Metal Poisoning

Metal	Key Features	Antidote	Classic Finding
Arsenic	Rice-water stools, garlic odor	BAL (dimercaprol)	Ideal homicidal poison
Lead	Wrist/foot drop, blue line	Ca-EDTA, succimer	Burton's line on gums
Mercury	Stomatitis, tremors	BAL, penicillamine	Hatters shakes / erethism
Copper	Blue-green vomit	Penicillamine	Blue vitriol (CuSO4)

Snake Bite — Big 4 of India

Snake	Venom Type	Key Feature	Treatment
Cobra (Naja)	Neurotoxic	Resp. paralysis	Anti-snake venom (ASV)
Krait (Bungarus)	Neurotoxic	Abdominal pain, night bite	ASV + neostigmine
Russell's Viper	Vasculotoxic	DIC, bleeding, renal failure	ASV (polyvalent)
Saw-scaled Viper	Vasculotoxic	Local swelling, bleeding	ASV, blood products

Dhatura = roadside robbery poison! Antidote table is MUST for exams!

Important Plant + Drug Poisons

Poison	Source / Type	Key Features
Dhatura	Anticholinergic plant	Hot as hare, blind as bat, dry as bone, mad as hatter
Oleander	Cardiac glycoside	Yellow oleander (Cerbera) = commonest suicidal plant poison
Abrus precatorius	Toxalbumin	Jequirity bean, 1 seed can be fatal, used in Sui poison
Opium / Morphine	CNS depressant	Pin-point pupils, resp. depression, needle marks
Barbiturates	Sedative-hypnotic	Drowsiness -> coma -> resp. failure, NO specific antidote
Cyanide	Asphyxiant	Bitter almond smell, cherry-red color, rapid death
Strychnine	Spinal stimulant	Opisthotonus, risus sardonicus, death in convulsions
Aconite	CVS poison	Tingling/numbness, called Queen of poisons

★ **Dhatura Mnemonic (Anticholinergic)**
Hot as a Hare, Blind as a Bat, Dry as a Bone, Red as a Beet, Mad as a Hatter
Treatment: Physostigmine (cholinesterase inhibitor) — reverses anticholinergic effects

Common Antidotes — MUST KNOW Table

Poison	Antidote	Mnemonic/Note
Organophosphorus	Atropine + PAM	PAM before aging (24h)
Benzodiazepines	Flumazenil	Fluma-ZENIL for benzo
Opioids / Morphine	Naloxone (Narcan)	Nal-OXONE for opioid
Paracetamol	N-Acetylcysteine (NAC)	NAC within 8-10 hours
Iron	Desferrioxamine	DES-ferro for iron
Arsenic / Mercury	BAL (Dimercaprol)	BAL = British Anti-Lewisite
Lead	CaNa2-EDTA / Succimer	EDTA chelates lead
Cyanide	Amyl nitrite + Na thiosulfate	Nitrite + thiosulfate kit
Methanol	Ethanol / Fomepizole	Ethanol blocks ADH
Heparin	Protamine sulfate	Prot-AMINE for heparin
Warfarin	Vitamin K	Vit K for warfarin
Copper / Wilson's	Penicillamine	Penicill-AMINE chelates Cu

Toxicology = guaranteed
20+ marks in FMT!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Describe OP poisoning — mechanism, features, treatment, PM findings
- 2. Classify poisons. Describe corrosive acid poisoning (H_2SO_4)
- 3. Describe arsenic poisoning — why ideal homicidal poison? PM findings
- 4. Methyl alcohol poisoning — features, treatment, differentiate from ethanol
- 5. Describe snake bite management. Name Big 4 snakes of India

Most Repeated SAQs (5–6 marks)

- 1. DUMBELS mnemonic and muscarinic vs nicotinic effects
- 2. CO poisoning — mechanism, PM findings, cherry-red lividity
- 3. Dhatura poisoning — features and medico-legal importance
- 4. IPC sections related to poisoning (Sec 284, 328)
- 5. Antidote of OP / cyanide / methanol / arsenic (table!)
- 6. Preservation of viscera for chemical analysis

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Ideal homicidal poison	Arsenic (tasteless, odorless)
King of poisons / Poison of kings	Arsenic
Queen of poisons	Aconite
Bitter almond smell	Cyanide (HCN)
Garlic odor in poisoning	Arsenic / Phosphorus
Cherry-red lividity	CO poisoning (COHb)
Pin-point pupils	Opioid / OP poisoning
Rice-water stools in poisoning	Arsenic
Roadside robbery poison	Dhatura
Snowstorm vision / blindness	Methanol poisoning
Burton's line on gums	Lead poisoning
Most common suicidal plant poison	Yellow oleander (Cerbera)

High Yield Checklist

MUST KNOW

- OP mechanism & Rx
- DUMBELS mnemonic
- Antidotes table
- Arsenic features
- IPC sections

HIGH YIELD

- CO — cherry-red PM
- Methanol vs Ethanol
- Big 4 snakes India
- Corrosive acids table
- Dhatura features

BONUS

- Strychnine features
- Cyanide smell & Rx
- Viscera preservation
- Aconite — Queen
- Barbiturate overdose

♥ Every life you save from poisoning matters, Sammy! ♥